

Anthony Tong

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EDUCATION

University of California, San Diego

B.S. Computer Science, B.S. Applied Mathematics | GPA: 4.0

La Jolla, CA

Class of 2027

HONORS & AWARDS

- Regents Scholarship @ University of California, San Diego (Fall 2023 – Spring 2027)
- NSF REU: Interdisciplinary AI @ University of California, San Diego (Fall 2025 – Spring 2026)
- NSF REU: AI SUSTAIN @ University of Nevada, Las Vegas (Summer 2024)

PUBLICATIONS

SPECTRE: A Multimodal Spectral Transformer for Small Molecule Annotation

Journal of Chemical Information and Modeling

2026

Wangdong Xu*, Byeol Ryu*, **Anthony Tong**, Huanru Mao, Hyunwoo Kim, James Zhao, Chen Zhang, Yiran Xu, Mingxun Wang, William Gerwick, Garrison W. Cottrell

RESEARCH

GURU | Python, PyTorch

La Jolla, CA

Applied AI Research w/ Prof. Garrison W. Cottrell

Apr 2025 - Present

- Developing AI models for assisting structural elucidation in natural products
- Mentored 6 students from local San Diego colleges as part of the REU program

ESB.AI Lab | Python, OpenCV, Label Studio, PyTorch

La Jolla, CA

Applied AI/Computer Vision Research w/ Prof. Edwin A. Solares

Oct 2023 - Apr 2025

- Directed 4 groups consisting of over 15 people in total on various research projects
- *LLMs for Language Preservation*: developing Tesseract and LLM models for endangered languages, including Hawaiian and Sanskrit
- *A-Droids*: streamlining data collection for avocado tree phenotyping via autonomous drones
- *AvoVision*: training CV models for avocado tree identification and phenotyping
- *AbaloneVision*: training CV models for abalone sex classification and gonad segmentation

AI SUSTAIN REU | Python, PyTorch

Remote | Las Vegas, NV

Visiting Researcher @ UNLV w/ Prof. Shengjie Zhai, Dr. Ming Zhu

Jun 2024 - Apr 2025

- Studied vocabulary replacement in tokenization on LLM medical domain adaptation
- Demonstrated >5% improvement on GPT-2 PubMedQA benchmark

Regents Scholars Research Initiative | Python, sklearn, cvxpy

La Jolla, CA

Machine Learning/AI Research w/ Prof. Arya Mazumdar

Sep 2023 - Jun 2024

- Worked on a variety of research topics including nonlinear estimation, generative models, and the performance of model mismatch
- Conducted experiments to validate the effect of temperature on sample complexity in logistic regression
- Investigated quantization algorithms for generative models and staircase/hierarchical learning

TEACHING

CSE 151B: Deep Learning

Prof. Qi (Rose) Yu | Head Tutor | SP26

La Jolla, CA

Apr 2026 — Jun 2026

CSE 150A: Intro to AI: Prob. Reasoning and Decision Making

Prof. Edwin A. Solares | Head Tutor | WI26

La Jolla, CA

Jan 2026 — Mar 2026

CSE 151A: Machine Learning: Learning Algorithms

Prof. Sanjoy Dasgupta | Tutor | FA25

La Jolla, CA

Sep 2025 — Dec 2025

CSE 150A: Intro to AI: Prob. Reasoning and Decision Making

Prof. Edwin A. Solares | Head Tutor | S125

La Jolla, CA

Jun 2025 — Jul 2025

CSE 151B: Deep Learning

Prof. Qi (Rose) Yu | Head Tutor | SP25

La Jolla, CA

Apr 2025 — Jun 2025

CSE 150A: Intro to AI: Prob. Reasoning and Decision Making

Prof. Edwin A. Solares | Head Tutor | WI25

La Jolla, CA

Jan 2025 — Mar 2025

INTERNSHIPS

Ridge Security Technology Inc. | Python

Remote | Fremont, CA

Returning Intern

Oct 2023 — Jan 2024

- Developed 6 plugins which integrate exploit scripts into the RidgeBot software

Summer Intern

Jun 2022 — Aug 2022

- Investigated 2 critical Windows Active Directory CVEs and created proof-of-concept (PoC) scripts